

C64 Keyboard (Hot Swap) Rev. 2

Bill of Material Rev. 2.0

Pos.	Qty	Value	Footprint	Ref.-No.	Comment	Group
1	1	225-2-01-01	2 Layer	_PCB Rev. 1	2 layer, Cu 35 μ , HASL, 385.6mm x 96.4mm, 1.6mm FR4	
2	4	100n	0805	C1, C2, C3, C4	Capacitor SMD 0805, 25V, X7R, $\pm 10\%$ or better	SL
3	1	1000u/10V	C30X16L	C4	electrolytic capacitor, axial leads, maximum length 30mm, maximum $\varnothing$ 16mm. E.g. Reichelt AX 105 1000/10	RGB
4	1	100n	0805	C5	Capacitor SMD 0805, 25V, X7R, $\pm 10\%$ or better	SL
5	2	100n	0805	C6, C8	Capacitor SMD 0805, 25V, X7R, $\pm 10\%$ or better	POR
6	1	4u7		C7	Capacitor SMD 0805, 25V, X7R, $\pm 10\%$ or better	POR
7	1	LL4148	SOD80C	D1	Diode, SMD	SL
8	2	LL4148	SOD80C	D2	Diode, SMD	SPK
9	1	SN74HC74D	SO-14	IC1	SMD IC, e.G. TI	SL
10	1	74HC123D	SO-16	IC2	SMD IC, e.G. TI	SL
11	1	TL7705ACD	SO8	IC3	SMD IC, e.G. TI	POR
12	1	box pin header, 2x10, 90°	2X10WH	J1	standard 2x10 pin box pin header, e.g. Reichelt.de WSL 20W	
13	1	pin header, 5p/90°, 2.54mm	1X05/90	J2	standard 90° pin header, e.g. Reichelt BKL 10120522	KSW
14	1	pin header, 5p/90°, 2.54mm	1X05/90	J6	DO NOT PLACE	
15	2	KF2510 , 3p/90°, 2.54mm	1X03/90	J3, J8	KF2510 (AliExpress) or Molex KK 2.54 (3p, 90°)	
16	1	pin header, 2p/90°, 2.54mm	1X02/90	J4	standard 90° pin header, e.g. Reichelt BKL 10120522	KSW

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17	1	Jumper, 2.54mm		(J4)	standard 2.54mm jumper, only required for short boards	KSW
18	1	pin header, 4p/90°, 2.54mm	1X04/90	J5	standard 90° pin header, e.g. Reichelt W+P 946-144	RGB
19	1	pin header, 4p/90°, 2.54mm	1X04/90	J7	standard 90° pin header, e.g. Reichelt W+P 946-144	I²C
20	1	PCF8575-MODULE	PCF8575MODU L	M1	Ali Express PCF8575 module. Read module description. Configuration of solder bridges requires	USBKB
21	2	10 way socket for pin header 2.54mm		(M1)	precision sockets, 2.54mm, 1x10, straight. E.g. Reichelt MPE 115-1-010	USBKB
22	1	5 way socket for pin header 2.54mm		(M1)	precision sockets, 2.54mm, 1x5, straight. E.g. Reichelt MPE 115-1-005	USBKB
23	1	5 way pin header, 2.54mm		(M1)	straight, e.g. Reichelt BKL 10120503	USBKB
24	1	PRO_MICRO	PRO_MICRO	M2	Ali Express pro micro module. USB-C connector preferred. Read module description	CTRL
25	2	12 way socket for pin header 2.54mm		(M2)	precision sockets, 2.54mm, 1x12, straight. E.g. Reichelt MPE 115-1-012	CTRL
26	3	BC847B	SOT23	Q1, Q3, Q5		
27	1	100R*	0805	R1	Resistor SMD 0805, 5% or better. Read Module Description	SL
28	1	330R	0805	R11	Resistor SMD 0805, 5% or better	
29	1	470R	0805	R16	Resistor SMD 0805, 5% or better	RGB
30	1	4k7	0805	R18	Resistor SMD 0805, 5% or better	SPK
31	1	33R	0805	R19	Resistor SMD 0805, 5% or better	SPK
32	11	10k	0805	R2, R3, R5, R7	Resistor SMD 0805, 5% or better	SL
33	11	10k	0805	R8, R9, R10, R12, R13, R14, R15	Resistor SMD 0805, 5% or better	

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34	1 27k	0805	R20	Resistor SMD 0805, 5% or better	SPK
35	1 27k	0805	R4	Resistor SMD 0805, 5% or better	SL
36	1 dnp	0805	R6	DO NOT PLACE	
37	1 47k	RN-9	RN1	Resistor network, 8 resistors/9 pins. E.g. USBKB Reichelt SIL 9-8 47K	
38	1 RMP-14P/HT	RMP-14P/HT	SPK1	piezzo buzzer, Ø14mm, pin distance 7.6mm, EKULIT (Reichelt SUMMER EPM 121)	SPK
39	58 Cherry MX compatible switch, 5 Pin	CHERRYMX	SW1, SW2, SW3, SW4, SW5, SW6, SW7, SW8, SW9, SW10, SW11, SW12, SW13, SW14, SW15, SW16, SW19, SW20, SW21, SW22, SW23, SW24, SW25, SW26, SW27, SW28, SW29, SW30, SW31, SW34, SW35, SW36, SW37, SW38, SW39, SW40, SW41, SW42, SW43, SW44, SW45, SW46, SW47, SW48, SW50, SW52, SW53, SW54, SW55, SW56, SW57, SW58, SW59, SW60, SW61, SW63, SW64, SW66	Cherry MX compatible switches of your choice. Read the module description.	
40	8 Cherry MX compatible switch, 5 Pin	CHERRYMX	SW17 or SW17A, SW18 or SW18A, SW32 or SW32A, SW33 or SW33A, SW49 or SW49A, SW51 or SW51A, SW62 or SW62A, SW65 or SW65A	SW** = for original keycaps, SW**A = for modern keycaps. Read module description	
41	2 Cherry MX compatible switch, 5 Pin	CHERRYMX	SW67, SW68		KSW and RGB

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42	76	SMD Hot Swap Sockets		SW1-68, SW17A, SW18A, SW32A, SW33A, SW49A, SW51A, SW62A, SW65A	E.g. Kailh CPG151101S11-16 (AliExpress). Read module description	
43	1	LED (high brightness)	LED 3mm	(SW35)	e.G. Kingbright L-7104SRC-J4 (Reichelt LED 3-4500 RT)	SL
44	1	Stabilizer, screw in type 7U		(SW66)	(for 7U space bar or 9U spacebar of C64U/Blingboard64 (CBMStuff.com). Read module description)	
45	1	Stabilizer, screw in type 2U		(SW48)	for RETURN key	
46	58	1U, blank keycaps			for MX compatibel switches, DSA profile. Read module description	
47	8	1.5U, blank keycaps			for MX compatibel switches, DSA profile. Read module description	
48	1	2U, blank keycap		(SW48)	for MX compatibel switches, DSA profile. Read module description	
49	1	7U space bar		(SW66)	for MX compatibel switches, DSA profile. Read module description	
50	25	C 2.2 x 6,5 H			screw, self tapping, DIN 7981	
51	2	C 2.9 x 9,5 H			screw, self tapping, DIN 7981	
52	2	3.2mm washer				
53	1	3D printed keyboard frame			Rev. 1	
54	1	ribbon cable + connector PCB			Project No. 224	
55	1	kernal switcher cable			Drawing 223-3-001-00. Read module description	KSW
56	1	optional WS2812B RGB LED strip			Read module description	RGB
57	1	optional OLED display			Read module description	I ² C

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Grp.	SL	Shift Lock Circuit				
	POR	Power on reset				
	RGB	RGB LED strip				
	SPK	Piezo Buzzer				
	KSW	Kernal Switcher				
	USBKB	USB Keyboard				
	I ² C	I ² C (OLED) Display				
	CTRL	Controller/pro micro				